

1995 HKCEE MATHS Paper II

1 Round off the number 0.044449 to 3 significant figures.

- A. 0.04
B. 0.044
C. 0.045
D. 0.0444
E. 0.0445

96 $\frac{1}{2+\sqrt{6}} - \frac{1}{2-\sqrt{6}} =$

- A. $-\sqrt{6}$
B. $-\frac{\sqrt{6}}{2}$
C. 0
D. $\frac{\sqrt{6}}{2}$
E. $\sqrt{6}$

2 If $\frac{x+y}{xy} = 1$, then $y =$

- A. $\frac{1-x}{x}$
B. $\frac{x-1}{x}$
C. $\frac{x}{1-x}$
D. $\frac{x}{x-1}$
E. $\frac{1-x}{1+x}$

97 The L.C.M. of $x^3 - x$ and $x^4 - 1$ is

- A. $x-1$
B. $(x-1)(x+1)$
C. $x(x-1)(x+1)(x^2+1)$
D. $(x-1)(x+1)(x^2+1)(x^3+x+1)$
E. $x(x-1)^2(x+1)^2(x^2+1)$

3 If $f(x) = x^{99} + 99x + k$ is divisible by $x+1$, then $k =$

- A. -100
B. -98
C. 98
D. 100
E. 98

98 Solve the simultaneous equations: $\begin{cases} 4x - \frac{y}{3} = 6 \\ 2x + \frac{y}{6} = -1 \end{cases}$

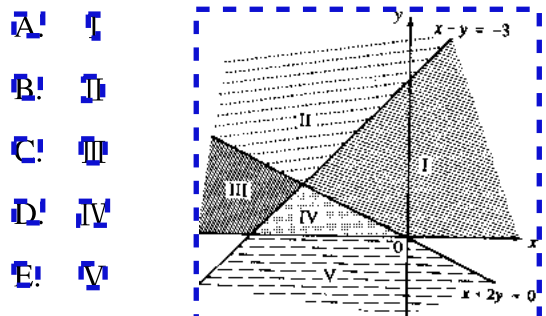
- A. $x = -\frac{1}{2}, y = -12$
B. $x = -\frac{1}{2}, y = 12$
C. $x = \frac{1}{2}, y = -12$
D. $x = \frac{1}{2}, y = 12$
E. $x = \frac{5}{24}, y = -\frac{7}{2}$

4 Simplify $\left(\frac{b^6}{b^{12}}\right)^{-\frac{2}{3}}$

- A. $\frac{b^8}{a^4}$
B. $\frac{b^{18}}{a^9}$
C. $\frac{a^4}{b^8}$
D. $\frac{a^9}{b^{18}}$
E. $\frac{1}{a^4 b^{12}}$

99 Which of the following shaded regions represents the solution of $\begin{cases} y \geq 0 \\ x - y \geq -3 \\ x + 2y \leq 0 \end{cases}$

represents the solution of $\begin{cases} y \geq 0 \\ x - y \geq -3 \\ x + 2y \leq 0 \end{cases}$



Label 9 Find the values of x which satisfy both

$$\text{and } \frac{2x-16}{3} > -2$$

- A. $-4 < x < 5$
- B. $x < -4$
- C. $x > -4$
- D. $x < 5$
- E. $x > 5$

Label 10 If $3x^2 + 6x + 1 \equiv 3(x + b)^2 + c$, then $c =$

- A. -8
- B. -2
- C. 0
- D. $\frac{1}{3}$
- E. 1

Label 11 x and y are two variables. The table below shows some values of x and their corresponding values of y .

x	36	16	4	12
y	36	16	4	1

Which of the following may be a relation between x and y ?

- A. $x \propto \sqrt{y}$
- B. $x \propto y$
- C. $x \propto \frac{1}{\sqrt{y}}$
- D. $x \propto \frac{1}{y}$
- E. $x \propto \frac{1}{y^2}$

Label 12 If $125^x = 25^y$ and x, y are non-zero, find $x : y$

- A. $1 : 25$
- B. $1 : 5$
- C. $2 : 3$
- D. $3 : 2$
- E. $5 : 1$

Label 13 Find the interest on $\$P$ at $r\%$ p.a. for n years, compounded half-yearly.

- A. $\$P(1 + 2r\%)^n - \P

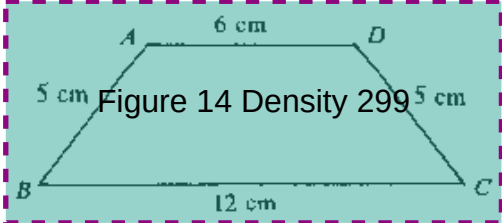
B. $\$P(1 + r\%)^n - \P

C. $\$P(1 + r\%)^{2n} - \P

D. $\$P\left(1 + \frac{r}{2}\%\right)^n - \P

E. $\$P\left(1 + \frac{r}{2}\%\right)^{2n} - \P

Label 14 In the figure, $ABCD$ is a trapezium. Find its area.

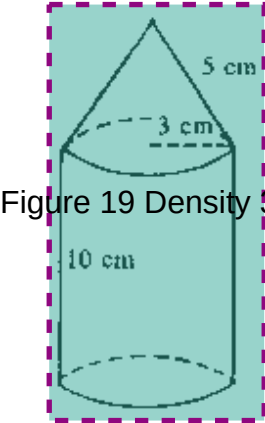


- A. 36cm^2
- B. 45cm^2
- C. 48cm^2
- D. 72cm^2
- E. 90cm^2

Label 15 In the figure, the solid consists of a cylinder and a right circular cone with a common base which

Label 18 has a radius of 3 cm. The height of the cylinder is 10 cm and the slant height of the cone is 5 cm. Find the total surface area of the solid.

- A. $75\pi\text{ cm}^2$
- B. $84\pi\text{ cm}^2$
- C. $93\pi\text{ cm}^2$
- D. $105\pi\text{ cm}^2$
- E. $114\pi\text{ cm}^2$



Label 16 $\frac{\cos^2 \theta}{1 + \sin \theta} - 1 =$

- A. $-\sin \theta$
- B. $\sin \theta$
- C. $\sin \theta - 2$
- D. $\frac{\sin \theta(1 - \sin \theta)}{1 + \sin \theta}$
- E. $\frac{\sin \theta(1 - \sin \theta)}{1 + \sin \theta}$

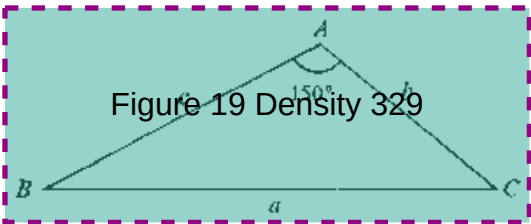
Label 17
 If $0 < x < 2\pi$, solve $\sin x = \frac{1}{3}$ correct to 3 significant figures!

- A. 0.327 or 2.81
- B. 0.327 or 3.47
- C. 0.340 or 2.80
- D. 0.340 or 3.48
- E. 0.340 or 5.94

Label 18
 The greatest value of $\frac{1}{2^{1-\sin x}}$ is

- A. $\frac{1}{2}$
- B. $\frac{1}{4}$
- C. 1
- D. 2
- E. 4

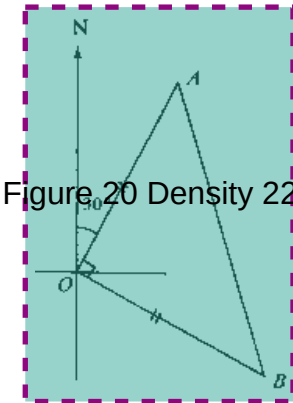
Label 19
 According to the figure, which of the following must be true?



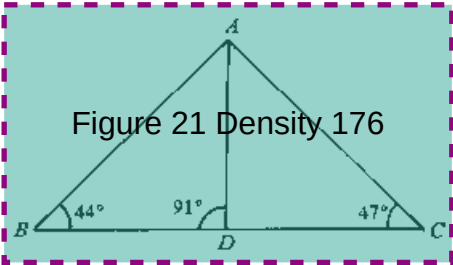
- A. $a^2 = b^2 + c^2 - \sqrt{3}bc$
- B. $a^2 = b^2 + c^2 - bc$
- C. $a^2 = b^2 + c^2 + \frac{\sqrt{3}}{2}bc$
- D. $a^2 = b^2 + c^2 + bc$
- E. $a^2 = b^2 + c^2 + \sqrt{3}bc$

Label 20
 In the figure, the bearing of B from A is

- A. 015°
- B. 045°
- C. 075°
- D. 165°
- E. 345°



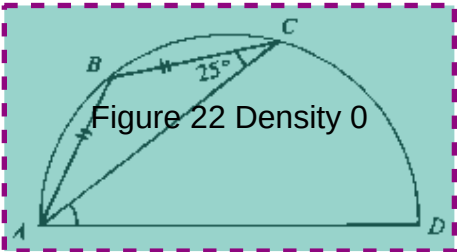
Label 21
 In the figure, BDC is a straight line. Arrange AD, BD and DC in ascending order of magnitude.



- A. $AD < BD < DC$
- B. $AD < DC < BD$
- C. $DC < AD < BD$
- D. $DC < BD < AD$
- E. $BD < AD < DC$

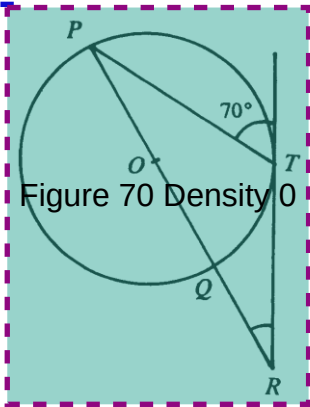
Label 22
 In the figure, ABCD is a semicircle. $\angle CAD =$

- A. 25°
- B. 40°
- C. 45°
- D. 50°
- E. 65°

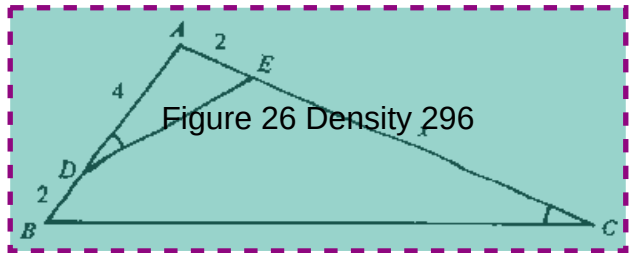


Label 23 In the figure, O is the center of the circle.
 $POQR$ is a straight line. TR is the tangent to the circle at T .
 Find $\angle PRT =$

- A. 20
- B. 35
- C. 45
- D. 50
- E. 70



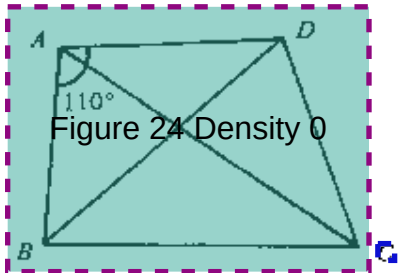
Label 26 In the figure, $\angle ADE = \angle ACB$. Find x .



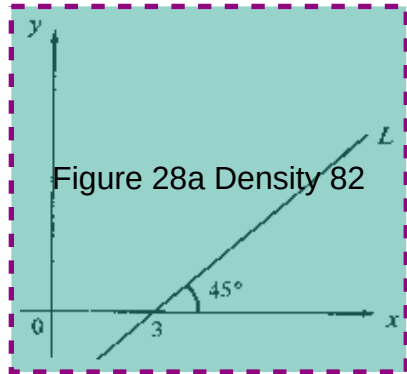
- A. 4
- B. 8
- C. 10
- D. 12
- E. 16

Label 24 In the figure, $ABCD$ is a cyclic quadrilateral. If $\angle DAB = 110^\circ$ and $BC \equiv BD$, find $\angle DAC$.

- A. 20°
- B. 35°
- C. 40°
- D. 55°
- E. 70°



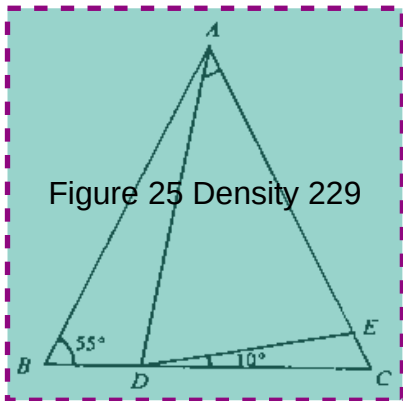
Label 27 In the figure, the equation of the straight line L is



- A. $x - 3 = 0$
- B. $x - y - 3 = 0$
- C. $x - y + 3 = 0$
- D. $x + y - 3 = 0$
- E. $x + y + 3 = 0$

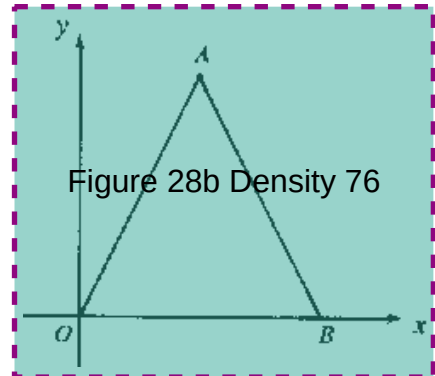
Label 25 In the figure, $AB = AC$ and $AD = AE$. $\angle DAC =$

- A. 45°
- B. 50°
- C. 55°
- D. 60°
- E. 65°



Label 28 In the figure, $OA = AB$. If the slope of AB is m , find the slope of OA .

- A. -1
- B. $\frac{1}{m}$
- C. $-\frac{1}{m}$
- D. m
- E. $-m$



Label 29 The table below shows the centers and radii of two circles C_1 and C_2 .

	Center	Radius
C_1	(2,2)	3
C_2	(5,-2)	2

Which of the following may represent the relative positions of C_1 and C_2 ?

- A

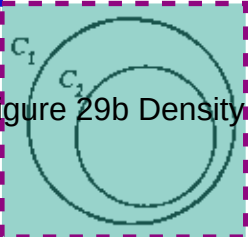


Figure 29b Density 0
- B

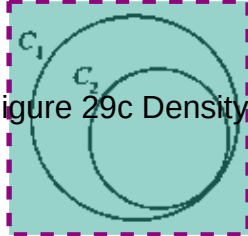


Figure 29c Density 0
- C

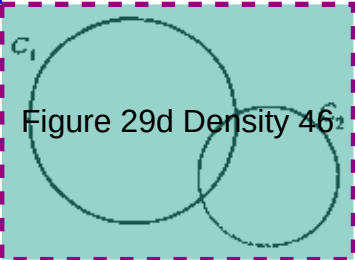


Figure 29d Density 462
- D

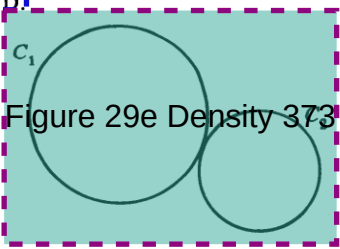


Figure 29e Density 373
- E

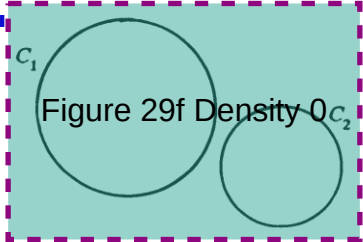
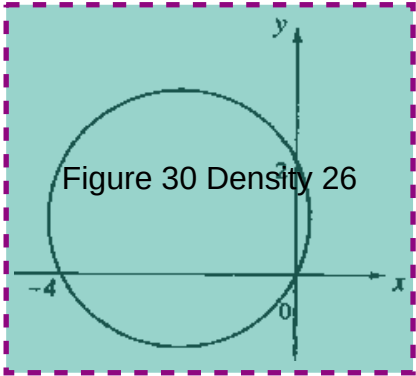


Figure 29f Density 0

Label 30 In the figure, the equation of the circle is

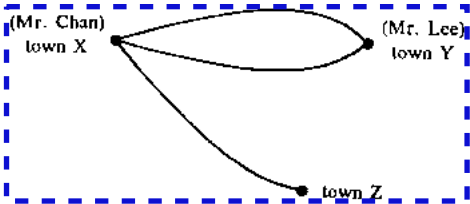


- A $x^2 + y^2 - 5 = 0$
- B $x^2 + y^2 - 2x + y = 0$
- C $x^2 + y^2 + 2x - y = 0$
- D $x^2 + y^2 - 4x + 2y = 0$
- E $x^2 + y^2 + 4x - 2y = 0$

Label 31 In a shooting game, the probability that A will hit a target is $\frac{3}{5}$ and the probability that B will hit it is $\frac{2}{3}$. If each fires once, what is the probability that they will both miss the target?

- A $\frac{1}{3}$
- B $\frac{1}{4}$
- C $\frac{2}{5}$
- D $\frac{2}{15}$
- E $\frac{11}{15}$

Label 32 The figure shows that Mr. Chan has 3 ways to leave town X and Mr. Lee has 2 ways to leave town Y. Mr. Chan and Mr. Lee leave town X and town Y respectively at the same time. If they select their ways randomly, find the probability that they will meet on their way.



A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. $\frac{1}{6}$

E. $\frac{5}{6}$

Label 33 The mean of a set of 9 numbers is 12. If the mean of the first 5 numbers is 8, the mean of the other four numbers is

A. 4

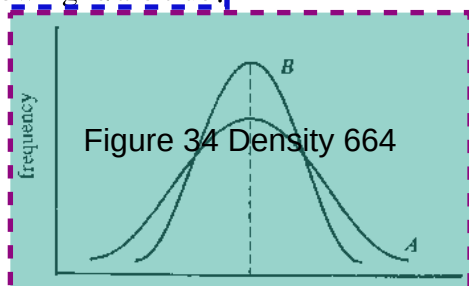
B. 10

C. 16

D. 17

E. 25

Label 34 The figure shows the frequency curves of two symmetric distributions A and B. Which of the following is/are true?



I. The mean of A = the mean of B

II. The inter-quartile range of A > the inter-quartile range of B

III. The standard deviation of A > the standard deviation of B

A. I only

B. I and II only

C. I and III only

D. II and III only

E. I, II and III

Label 35 If $f(x) = \frac{x}{1-x}$, then $f\left(\frac{1}{x}\right)f(-x) =$

A. $-\frac{1}{2}$

B. $-\frac{1}{3}$

D. $\frac{x}{1-x^2}$

C. $\frac{1-x}{1+x}$

E. $\frac{x}{x^2-1}$

Label 36 Label 37 $2a^{n+1} - 7a^n = 30a^{n-1}$

A. $(a^n - 6)(2a + 5)$

B. $a^n(a + 6)(2a - 5)$

C. $a^n(a - 6)(2a + 5)$

D. $a^{n-1}(a + 6)(2a - 5)$

E. $a^{n-1}(a - 6)(2a + 5)$

Label 37

Figure 38 Density 0

$$\frac{\left(\frac{y}{x} - 1\right)\left(1 - \frac{x}{y}\right)}{\frac{x}{x} - \frac{y}{y}}$$

A. $\frac{x-y}{x+y}$

B. $\frac{x-y}{x+y}$

C. $\frac{x+y}{x-y}$

D. $\frac{x+y}{x-y}$

E. $-\frac{1}{x}$

Label 38 If $5^a = 2^b = 10^c$ and a, b, c are non-zero,

then $\frac{c}{a} + \frac{c}{b} =$

A. $\frac{7}{10}$

B. 1

D. $\log 7$

C. 7

E. $\frac{1}{\log 2} + \frac{1}{\log 5}$

Label 39 α, β are the roots of the equation

Label 40 $x^2 - 4x - 3 = 0$, then $\alpha^2 + \alpha\beta + \beta^2 =$

Label 13

Label 14

D. 16

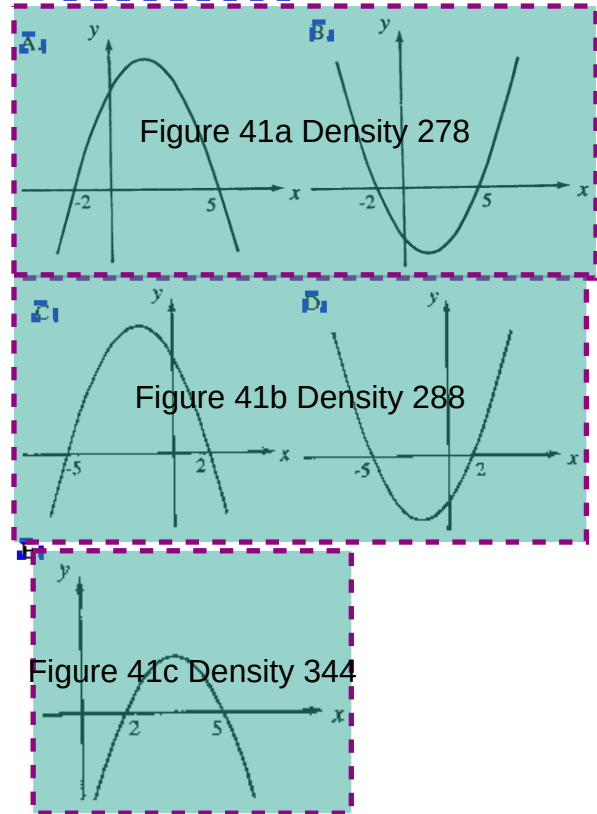
Label 13

E. 19

Label 40 Find the range of values of k such that the equation $x^2 + (k-2)x + 1 = 0$ has real roots.

- A. $k = 4$
- B. $0 < k < 4$
- C. $0 \leq k \leq 4$
- D. $k < 0$ or $k \geq 4$
- E. $k \leq 0$ or $k \geq 4$

Label 41 Which of the following may represent the graph of $y = -x^2 + 3x + 10$?



Label 42 In an A.S., the sum of the first 2 terms is 3 and the sum of the first 3 terms is 2. The common difference is

- A. $\frac{5}{3}$
- B. $-\frac{5}{3}$
- C. 1
- D. $\frac{5}{3}$
- E. $\frac{7}{3}$

Label 43 If the geometric mean of two positive numbers a and b is 10, then $\log a + \log b =$

- A. $\frac{1}{2}$

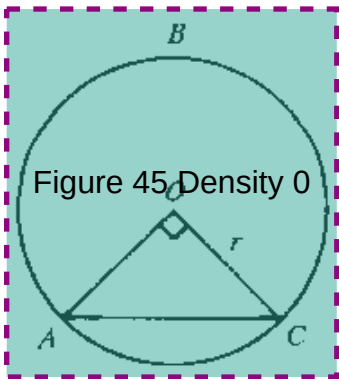
- B. 1
- C. 2
- D. 10
- E. 100

Label 44 The marked price of a toy is \$120 and the percentage profit is 60%. If the toy is sold at a discount of 20%, the profit is

- A. \$14.40
- B. \$21.00
- C. \$24.00
- D. \$33.60
- E. \$48.00

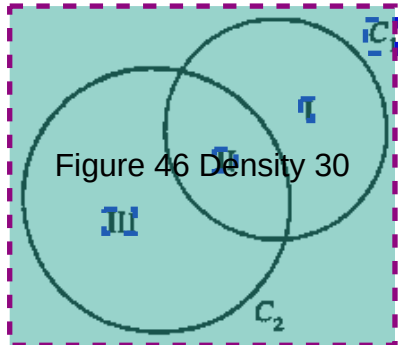
Label 45 In the figure, O is the center of the circle. Find the area of the major segment ABC .

- A. $\frac{\pi}{4}r^2$
- B. $\frac{3\pi}{4}r^2$
- C. $\left(\frac{\pi}{4} - \frac{1}{2}\right)r^2$
- D. $\left(\frac{3\pi}{4} - \frac{1}{2}\right)r^2$
- E. $\left(\frac{3\pi}{4} + \frac{1}{2}\right)r^2$



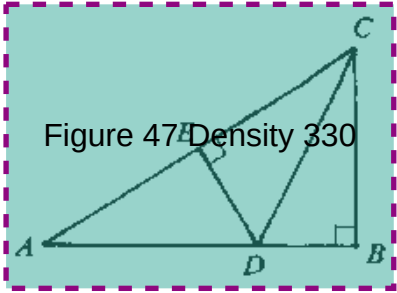
Label 46 In the figure, C_1 and C_2 are two circles. If area of region I : area of region II : area of region III $= 2 : 1 : 3$, then radius of C_1 : radius $C_2 =$

- A. $9 : 16$
- B. $2 : 3$
- C. $3 : 4$
- D. $\sqrt{2} : \sqrt{3}$
- E. $\sqrt{3} : 2$

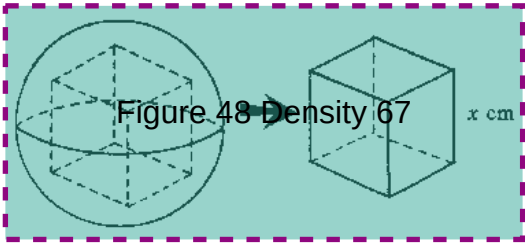


Label 47 In the figure, $DE=DB$, $AC=13$ and $BC=5$. Area of $\triangle ADE$: Area of $\triangle ACB=$

- A. 64 : 169
- B. 5 : 13
- C. 4 : 9
- D. 8 : 13
- E. 2 : 3



Label 48 In the figure, a solid wooden sphere of radius 3cm is to be cut into a cube of side x cm. Find the largest possible value of x .



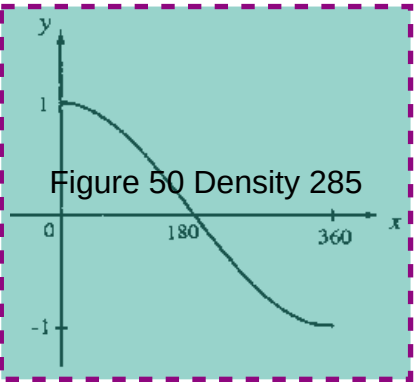
- A. $3\sqrt{2}$
- B. $2\sqrt{3}$
- C. 3
- D. $\frac{3}{2}\sqrt{2}$
- E. $\sqrt{3}$

Label 49 In $0^\circ \leq x \leq 360^\circ$, the number of points of intersection of the graphs $y = \sin x$ and $y = \tan x$ is

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

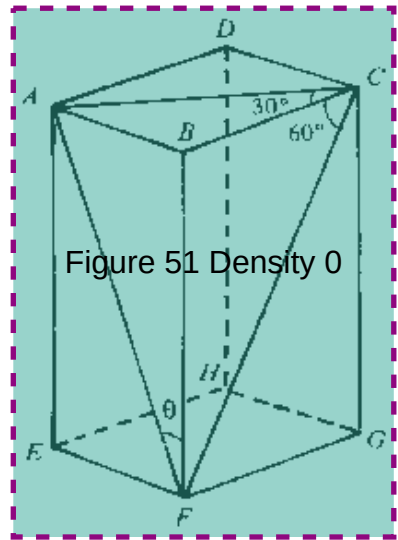
Label 50 The figure shows the graph of the function

- A. $y = \cos \frac{x^\circ}{2}$
- B. $y = \frac{1}{2} \cos x^\circ$
- C. $y = \cos x^\circ$
- D. $y = 2 \cos x^\circ$
- E. $y = \cos 2x^\circ$



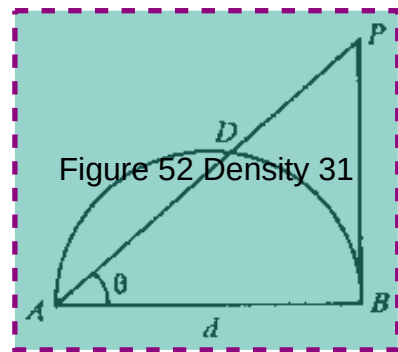
Label 51 In the figure, $ABCDEFGH$ is a cuboid, $\tan \theta =$

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. 1
- D. $\sqrt{3}$
- E. 3

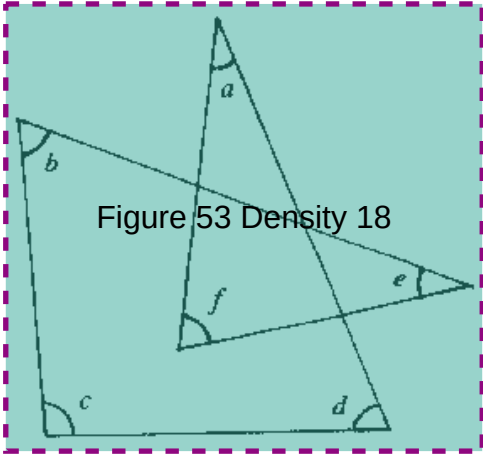


Label 52 In the figure, PB touches the semicircle ADB at B . $PD=$

- A. $\frac{d}{2 \cos \theta}$
- B. $d \sin \theta \tan \theta$
- C. $\frac{d}{\sin \theta \tan \theta}$
- D. $\frac{d \cos \theta}{\tan \theta}$
- E. $\frac{d \tan \theta}{\cos \theta}$

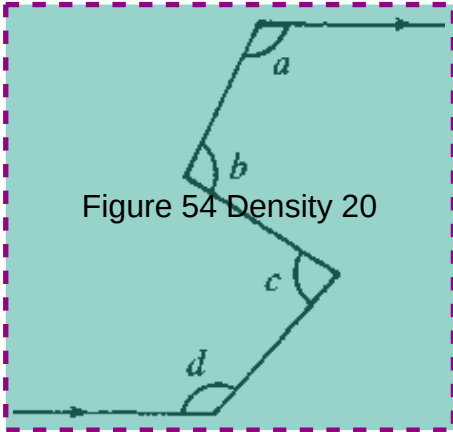


Label 53
 In the figure, $a + b + c + d + e + f =$



- A. 270°
- B. 360°
- C. 450°
- D. 540°
- E. 720°

Label 54
 According to the figure, which of the following must be true?



- A. $a + b = c + d$
- B. $a + d = b + c$
- C. $a + b + c + d = 360^\circ$
- D. $a + b + c + d = 540^\circ$
- E. $2a + 2b - c = d = 720^\circ$

END OF PAPER